

Approximate areas under graphs



1

- a
- i 31.193 units²
 - ii 84.791 units²
 - iii 51.483 units²
- b $e^4 - 1$ units²
- c 4%

2

- a
- i 8.418 units²
 - ii 9.331 units²
 - iii 8.928 units²
- b 8.8827 units²
- c 0.51%

3

- a
- i 1.239 units²
 - ii 1.149 units²
 - iii 1.016 units²
- b 1 unit²
- c 2%

4

- a $f'(x) = e^{-x} - xe^{-x}$
- b No solution provided.
- c
- i 1.034 units²
 - ii 0.861 units²
- d $\frac{2}{e} - \frac{6}{e^5}$ units²

e

	2 rectangles	4 rectangles
Percentage error	48.71	23.83

5

- a 40.5984 units²
- b 30.9605 units²
- c As $f(x)$ is convex over the interval $1 \leq x \leq 3$, an underestimate will include a smaller error in the calculation than an overestimate. Hence, the calculation in part (b) will be closer to the exact value.

6

- a 8.008 units²
- b 9.594 units²

- c** As $f(x)$ has a gradient closer to horizontal for $2 \leq x \leq 4$ than $0 \leq x \leq 2$, the rectangles used in the overestimate in part (b) are closer to the actual shape of the function. Hence, the calculation in part (b) includes a smaller error in the calculation and will be closer to the exact value.
- d** As $f(x)$ is concave over the interval $0 \leq x \leq 4$, an overestimate will include a smaller error in the calculation than an underestimate. Hence, an overestimate will be closer to the exact value.

Definite integration

7

a $\frac{16}{3} \text{ units}^2$

b $\frac{32}{3} \text{ units}^2$

c $1 : 2$

8 $1 : 7$

9 $k = \frac{\pi}{6}$

10 $B = A^2$

11

a $7\pi^2 + 2$

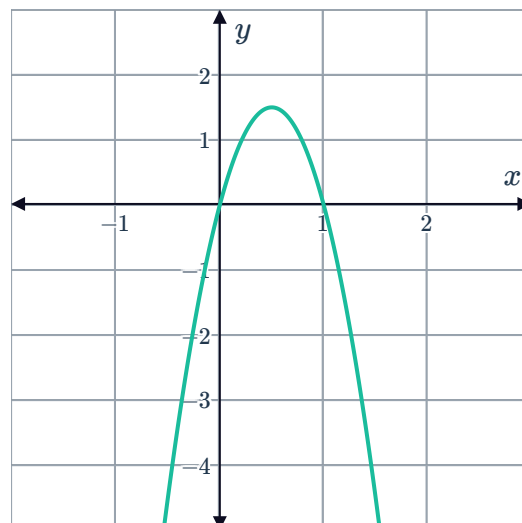
b $4\pi^2 + 2 \text{ units}^2$

c $12\pi^2 - 2 \text{ units}^2$

d $1 : 2.81$

12

a



b 1 unit^2

c	a	1	2	3	5	
	Area (units) ²	1	8	27	125	1000

- d The area bounded by $f(x) = 6ax - 6x^2$ and the x -axis, for $a \geq 0$ is a^3 units².
- e No solution provided.

13

a $k = 6$

b $\frac{1}{12}$ units²

c 12

d

n	1	2	3	4	5
k	6	12	20	30	42

- e For the area under the curve of $f(x) = kx^n(1 - x)$ on the interval $[0, 1]$ to be 1 unit², $k = (n + 1)(n + 2)$.
- f No solution provided.

14

a 1 unit²

b i $\cos 2x$ ii $0 \leq x \leq \frac{\pi}{4}$ iii $\frac{1}{2}$ units²

c $\frac{1}{3}$

- d The area bounded by $y = f_k(x)$ and the x -axis over $0 \leq x \leq \frac{\pi}{2k}$, for any positive integer value of k is $\frac{1}{k}$ units².
- e No solution provided.